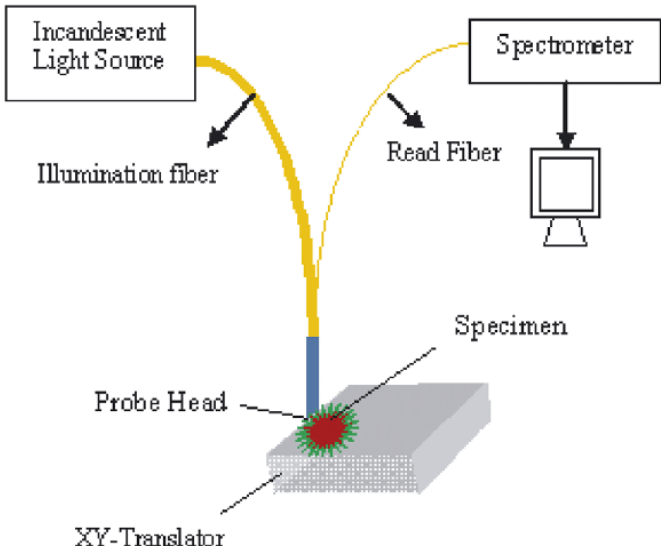
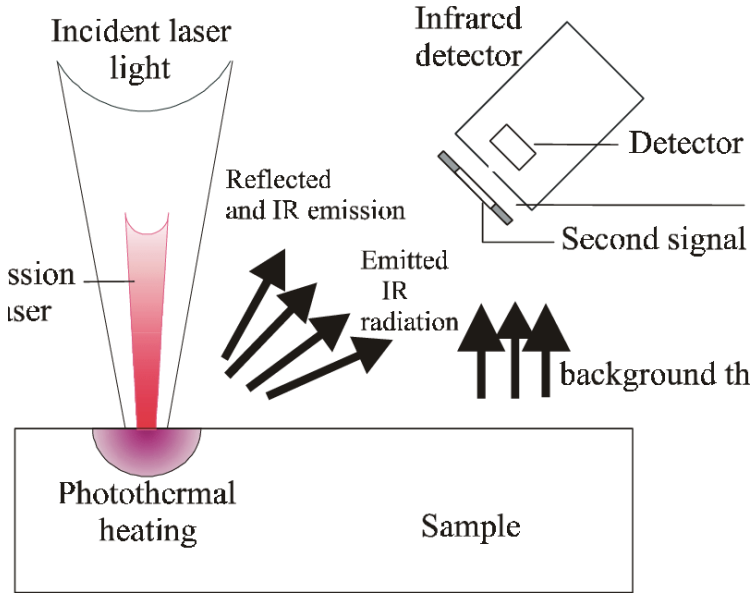
Nano Photonics and Plasmonics Research

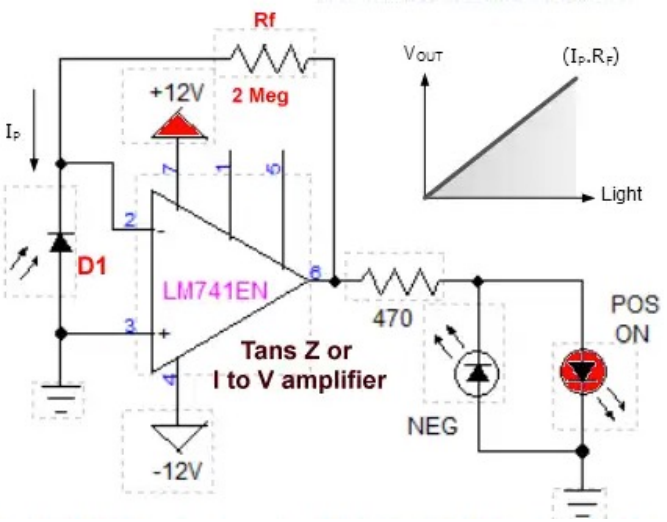
Overview Lecture

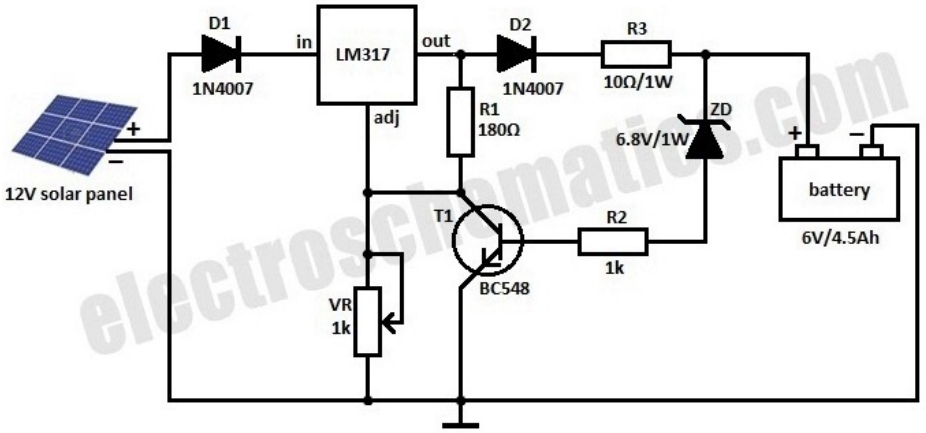
Opto-electronic Devices and Circuits

Optoelectronics

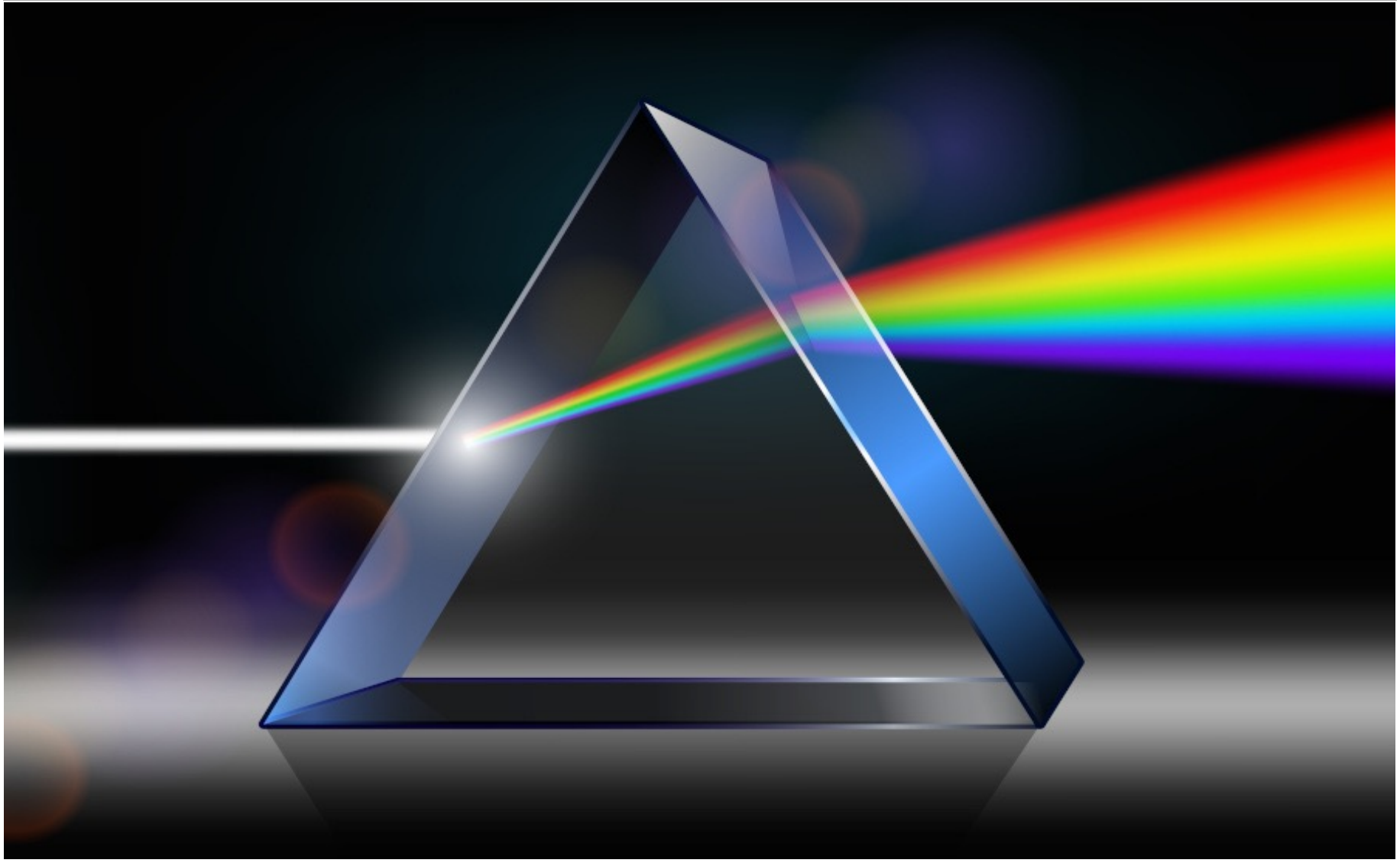




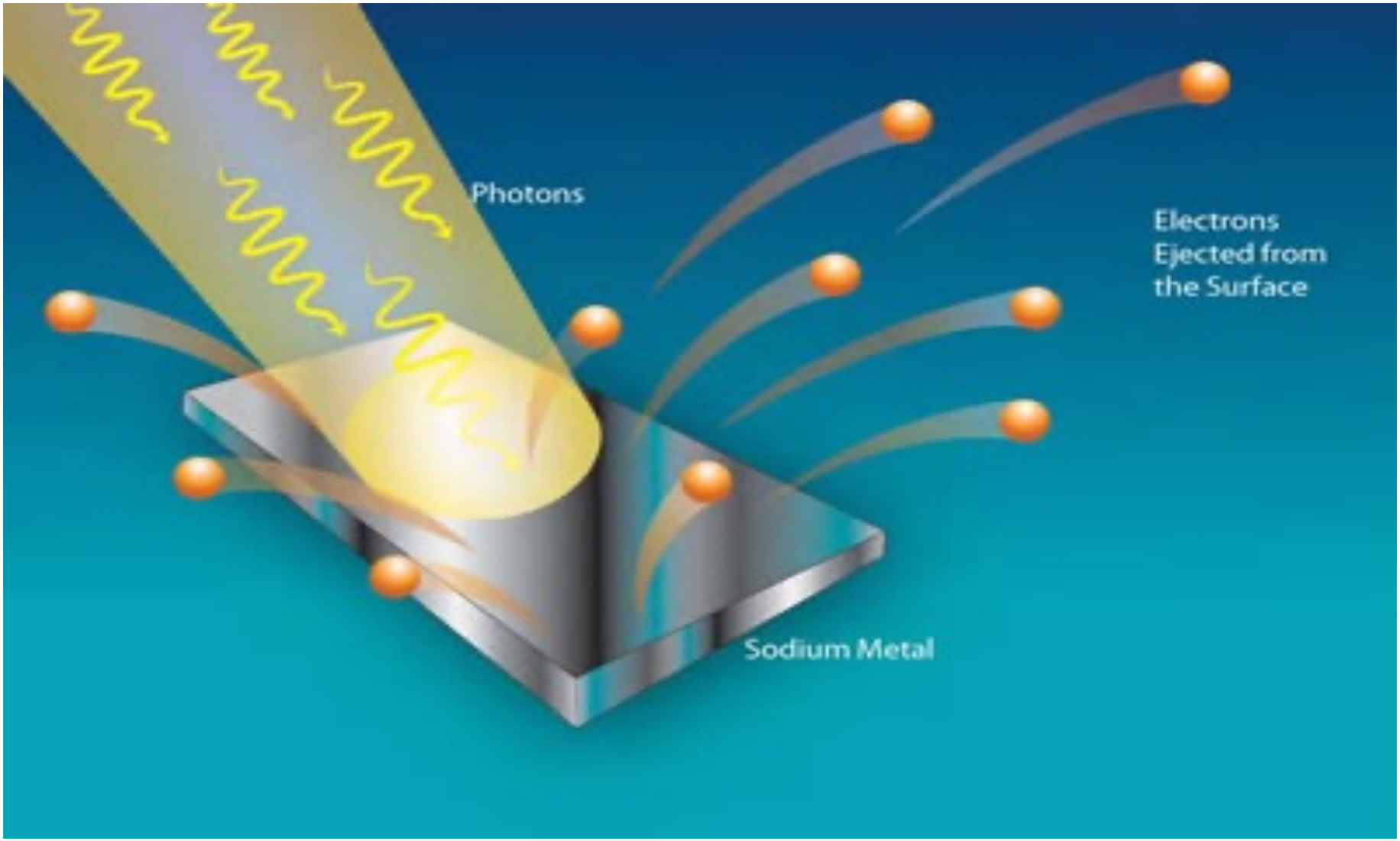




Wave Nature of Light



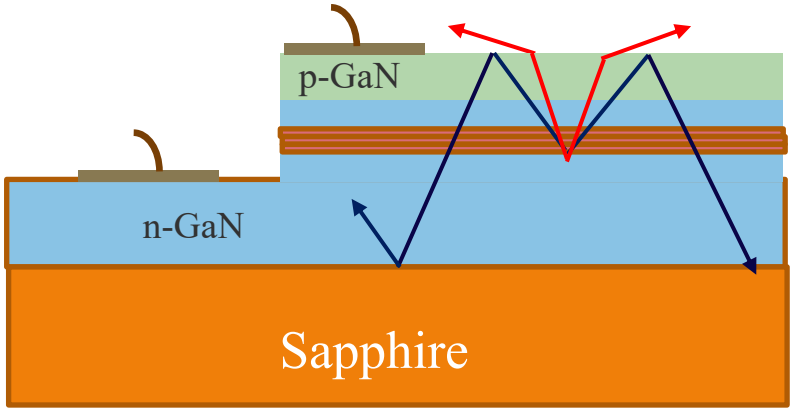
Particle Nature of Light

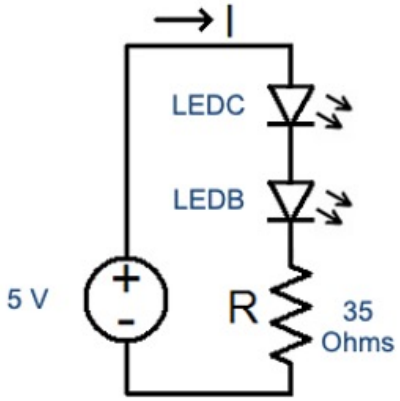


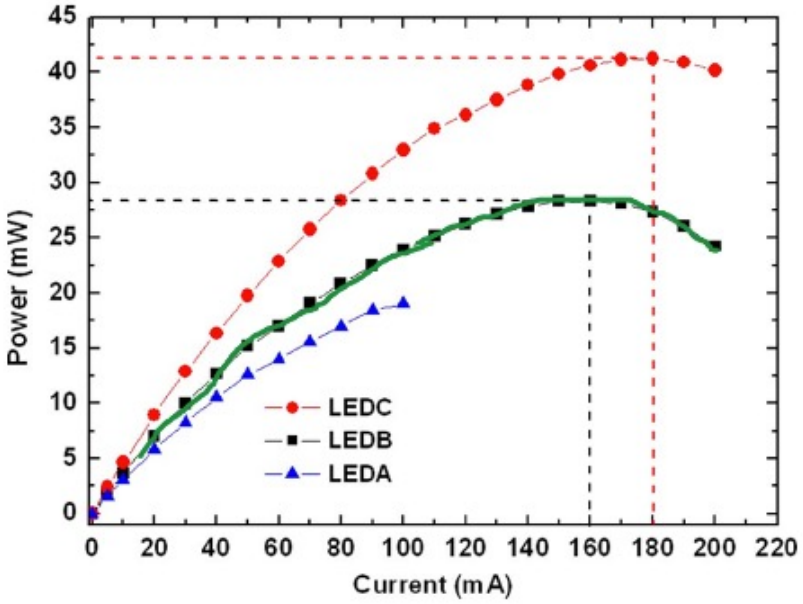
Semiconductors for Optoelectronics

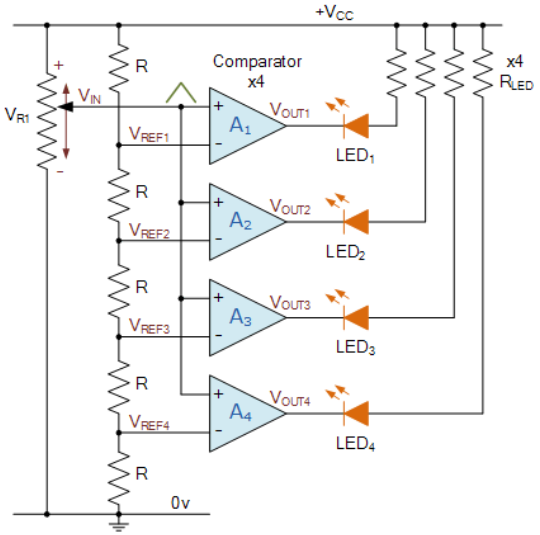


Light Emitting Devices

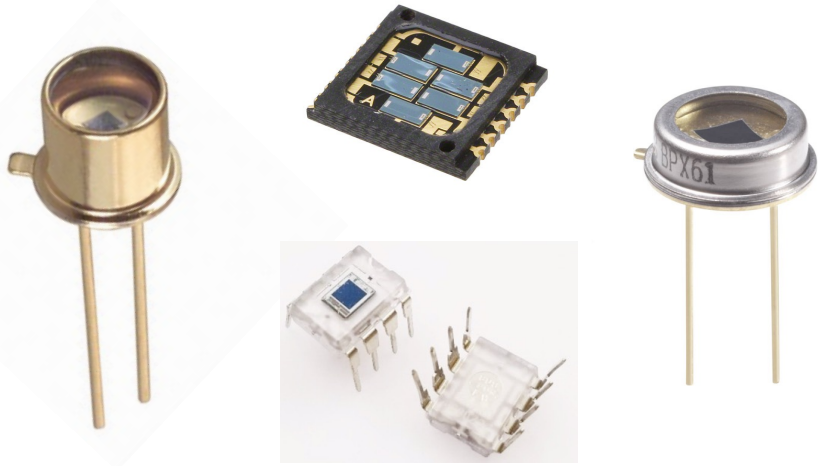


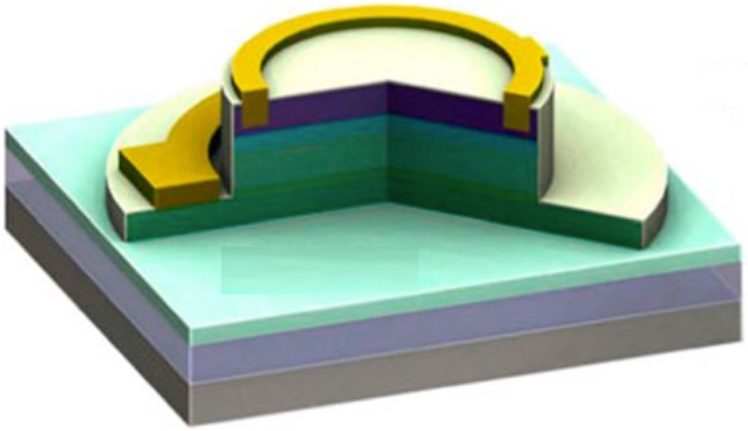




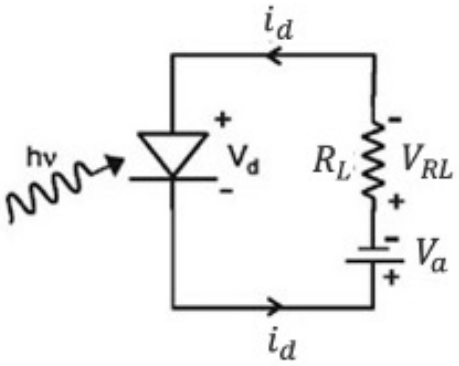


Photoconductors

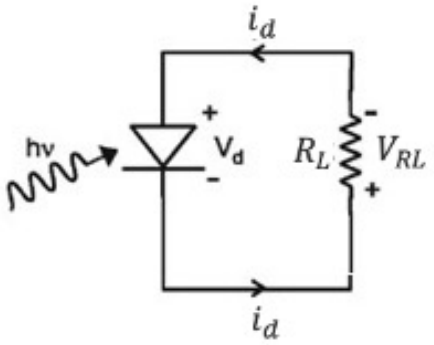




Photoconductive Mode



Photovoltaic Mode



Syllabus

Nano Photonics and Plasmonics Research

Instructor Information	Mohammad A. Alsunaidi	Office M4-232	Phone ---	Office Hours Mon-Thu: 11:30-12:30	Email mohammad.alsunaidi@marmara	
Course Information	Text *Optoelectronics*, S. O. Kasap	Grading	Class Work Assignments + Exp (4x5)% + Project 10%		Midterm Exam 30%	Final Exam 40%

Week	Events	Subjects	Tuesdays	Wednesdays
1 Oct 2-6	---	Nature of Light and Material Properties	Course introduction and scope.	Light as an EM Wave: Maxwell's and Wave Equations. Types of waves.
2 Oct 9-13	---		Light Propagation equation.	Light Propagation in different media.
3 Oct 16-20	Course drop/add Week		Light Reflection and Refraction.	Light Reflection and Refraction. Anti-reflection Surfaces.
4 Oct 23-27	Cumhuriyet Bayramı Oct 29		Light-matter interaction	Light as an energy quanta: The photoelectric effect and the Photon.
5 Oct 30-Nov 3	---	The PN Junction	Semiconductor Fundamentals and the PN Junction. PN junction energy bands.	Photon absorption and generation. Direct and Indirect semiconductors.
6 Nov 6-10	Assignment 1 Due Nov 10	Photodetectors	Photodetectors types and basic structures.	Characteristics and modes of operation. PN, PiN, APD and Schottky PD.
7 Nov 13-17	---		Photodiode circuit configurations and modes of operation.	Phototransistor circuit configurations and modes of operation.
Nov 20-26	**Mid-term Exams period**			
9 Nov 27-Dec 1	Experiment 1 Due Nov 17	Solar Cells	Basic and improved solar cells structures and materials. Thin film cells. Design rules.	Photo-voltaic (PV) circuit analysis.
10 Dec 4-8	---	Light Emitting Diodes (LEDs)	LEDs basic design and operation.	LED characterization: optical power, spectral distribution and modulation BW.
11 Dec 11-15	Assignment 2 Due Dec 15		LED characterization: optical power, spectral distribution and modulation BW.	Improved LED structures and materials.
12 Dec 18-22	Midterm Make-up (Excused, Dec 21-23)		UV LEDs. Material systems, basic designs and challenges.	White LEDs. Techniques, efficiencies and challenges.
13 Dec 25-29	Experiment 2 Due Dec 29		LED circuits. Basic amplifier configurations.	LED circuits. Modules, and panels.
14 Jan 1-5	New Year 2024 Monday	Opto-couplers and Opto-isolators	Opto-couplers basic design and applications.	Opto-couplers circuits.
15 Jan 8-12	Project submission Jan 12	Special topics	Students' Projects	Students' Projects
Jan 15-28	**Final Exams period**		**Jan 29-Feb 4 (Re-sit Final Exams period)**	

Subjects

Light: Wave nature, Particle nature (Photon)

Semiconductors

The PN junction

Photo-diodes and Circuits

LEDs and Circuits

Light as an Electromagnetic Wave

The following subjects will be covered

- Maxwell's Equations: Review
- The EM Wave Equation and Solutions.
- The Refractive Index Relation.
- Light Wave Power Density.

Maxwell's Equations

Review of the fundamental laws of electricity and magnetism.

Fields	$\vec{E} = \frac{\vec{F}_e}{q}$	$(\vec{v} \times \vec{B}) = \frac{\vec{F}_m}{q}$
Flux densities	$\vec{D} = \varepsilon \vec{E}$	$\vec{B} = \mu \vec{H}$
Gauss's Law	$\vec{\nabla} \cdot \vec{D} = \rho_v$	$\oint \vec{D} \cdot \overrightarrow{ds} = Q_{enc}$
Gauss's Law	$\nabla \cdot \vec{B} = 0$	$\oint \vec{B} \cdot \overrightarrow{ds} = 0$
Ampere's Law	$\vec{\nabla} \times \vec{H} = \vec{J}$	$\oint \vec{H} \cdot \overrightarrow{dl} = I_{enc}$
Faraday's Law	$\mathcal{E}mf = -\frac{\partial \varphi}{\partial t}$	$\vec{\nabla} \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$

Charge conservation
The continuity equation

$$\vec{\nabla} \cdot \vec{J} + \frac{\partial \rho_v}{\partial t} = 0$$

$$\vec{J} = \sigma \vec{E}$$

Basic Quantities

(V/m) $\vec{E}, \vec{D}$ (C/m^2)

(A/m) $\vec{H}, \vec{B}$ (Wb/m^2)

(A/m^2) $\vec{J}, \rho_v$ (C/m^3)

Material Properties

ε, μ, σ

(F/m) (H/m) (S/m)

Maxwell's Equations

$$\vec{\nabla} \times \vec{H} = \vec{J} + \frac{\partial \vec{D}}{\partial t} \qquad \vec{\nabla} \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$$

$$\vec{\nabla} \cdot \vec{D} = \rho_v \qquad \nabla \cdot \vec{B} = 0$$

$$\vec{D} = \varepsilon \vec{E} \qquad \vec{B} = \mu \vec{H} \qquad \vec{J} = \sigma \vec{E}$$

Maxwell's Equations

Constitutive Relations

$$\vec{\nabla} \times \vec{E} = -\frac{\partial \vec{B}}{\partial t} = -\mu \frac{\partial \vec{H}}{\partial t}$$

$$\vec{\nabla} \times \vec{\nabla} \times \vec{E} = -\mu \frac{\partial (\vec{\nabla} \times \vec{H})}{\partial t}$$

$$\vec{\nabla}(\vec{\nabla} \cdot \vec{E}) - \vec{\nabla}^2 \vec{E} = -\mu \frac{\partial}{\partial t}(\vec{J} + \frac{\partial \vec{D}}{\partial t})$$

$$\vec{\nabla}^2 \vec{E} - \vec{\nabla}(\vec{\nabla} \cdot \vec{E}) = \mu\sigma \frac{\partial \vec{E}}{\partial t} + \mu\varepsilon \frac{\partial^2 \vec{E}}{\partial t^2}$$

$$\vec{\nabla} \times \vec{\nabla} \times \vec{E} = \vec{\nabla}(\vec{\nabla} \cdot \vec{E}) - \vec{\nabla}^2 \vec{E}$$

$$\vec{\nabla}^2 \vec{E} = \mu\sigma \frac{\partial \vec{E}}{\partial t} + \mu\varepsilon \frac{\partial^2 \vec{E}}{\partial t^2}$$

$$\vec{\nabla}^2 \vec{H} = \mu\sigma \frac{\partial \vec{H}}{\partial t} + \mu\varepsilon \frac{\partial^2 \vec{H}}{\partial t^2}$$

The EM Wave Equation

The EM Wave Equation

$$\vec{\nabla}^2\vec{E} = \mu\sigma\frac{\partial\vec{E}}{\partial t} + \mu\varepsilon\frac{\partial^2\vec{E}}{\partial t^2} \qquad \vec{\nabla}^2\vec{H} = \mu\sigma\frac{\partial\vec{H}}{\partial t} + \mu\varepsilon\frac{\partial^2\vec{H}}{\partial t^2}$$

Consider a source-free $\rho_v = 0$ and lossless (non-conducting) $\sigma = 0$ medium

$$\vec{\nabla}^2\vec{E} = \mu\varepsilon\frac{\partial^2\vec{E}}{\partial t^2} \qquad \vec{\nabla}^2\vec{H} = \mu\varepsilon\frac{\partial^2\vec{H}}{\partial t^2}$$

Write the wave equation in this form

$$\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}\right)\vec{E} - \mu\varepsilon\frac{\partial^2\vec{E}}{\partial t^2} = 0 \qquad \nabla^2 U - \frac{1}{v^2}\frac{d^2U}{dt^2} = 0$$

Wave speed $v = \dfrac{1}{\sqrt{\mu\varepsilon}}$ For free space $v = \dfrac{1}{\sqrt{\mu_o\varepsilon_o}} = 3.0\times10^8$ (m/s)

1) Electromagnetic waves exist.

2) The wave is composed of two interacting quantities E and H fields.

3) EM waves travel with the speed of light.

4) Light is an EM waves.

TH Plane-wave Solution in Lossless Media

Consider a source-free $\rho_v = 0$ and lossless (non-conducting) $\sigma = 0$ medium

$$\vec{\nabla}^2\vec{E} = \mu\varepsilon\frac{\partial^2\vec{E}}{\partial t^2} \qquad \vec{\nabla}^2\vec{H} = \mu\varepsilon\frac{\partial^2\vec{H}}{\partial t^2}$$

Time-harmonic waves are generated by time-harmonic sources.

$\vec{E}(r,t) = \vec{E}(r)\,e^{j\omega t}$ r here represents the space variable. $e^{j\omega t}$ Sinusoidal, single frequency

$$\vec{\nabla}^2\vec{E}(r) + k^2\vec{E}(r) = 0$$
$$\vec{\nabla}^2\vec{H}(r) + k^2\vec{H}(r) = 0$$

Helmholtz wave equation

$$k = \omega\sqrt{\mu\varepsilon} = \frac{\omega}{v}$$

Wave number (rad/m) (Real number)

Refractive index

$$n = \sqrt{\mu_r\varepsilon_r} \qquad v = \frac{1}{\sqrt{\mu\varepsilon}} = \frac{c}{n} \qquad k = \frac{n\omega}{c}$$

The Uniform Plane-wave Solution

A general expression of the electric field vector in rectangular coordinates is

$$\vec{E}(x,y,z,t) = E_x(x,y,z,t)\,\vec{a}_x + E_y(x,y,z,t)\,\vec{a}_y + E_z(x,y,z,t)\,\vec{a}_z$$

TH Wave equation $\nabla^2\vec{E}(x,y,z) + k^2\vec{E}(x,y,z) = 0$ ➡ $\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2} + k^2\right)\vec{E}(x,y,z) = 0$

Consider a one-dimensional case (z) with one field component only.

$$\frac{\partial^2 E_x}{\partial z^2} + k^2 E_x = 0$$

$$E_x(z) = A\,e^{-jkz} + B\,e^{jkz}$$

Plane wave Solution

A and B are constants.

Put back the time-harmonic factor

$$E_x(z,t) = A\,e^{j(\omega t - kz)} + B\,e^{j(\omega t + kz)}$$

Wave travelling along the $+ve$ z direction

Wave travelling along the $-ve$ z direction

Take the $+z$ wave

$$E_x(z,t) = E_o\,e^{j(\omega t - kz)}$$

$$E_x(z,t) = E_o\cos(\omega t - kz) + j\,E_o\sin(\omega t - kz)$$

Solution Solution

Take the sine wave solution

$$E_x(z,t) = E_o\sin(\omega t - kz)$$

$$\vec{E}(z,t) = E_o\sin(\omega t - kz)\;\vec{a}_x$$

Same definitions for E_o, ω, k

Time-harmonic (Sinusoidal) Light Wave

Electric field Part of the wave

Amplitude

Phase

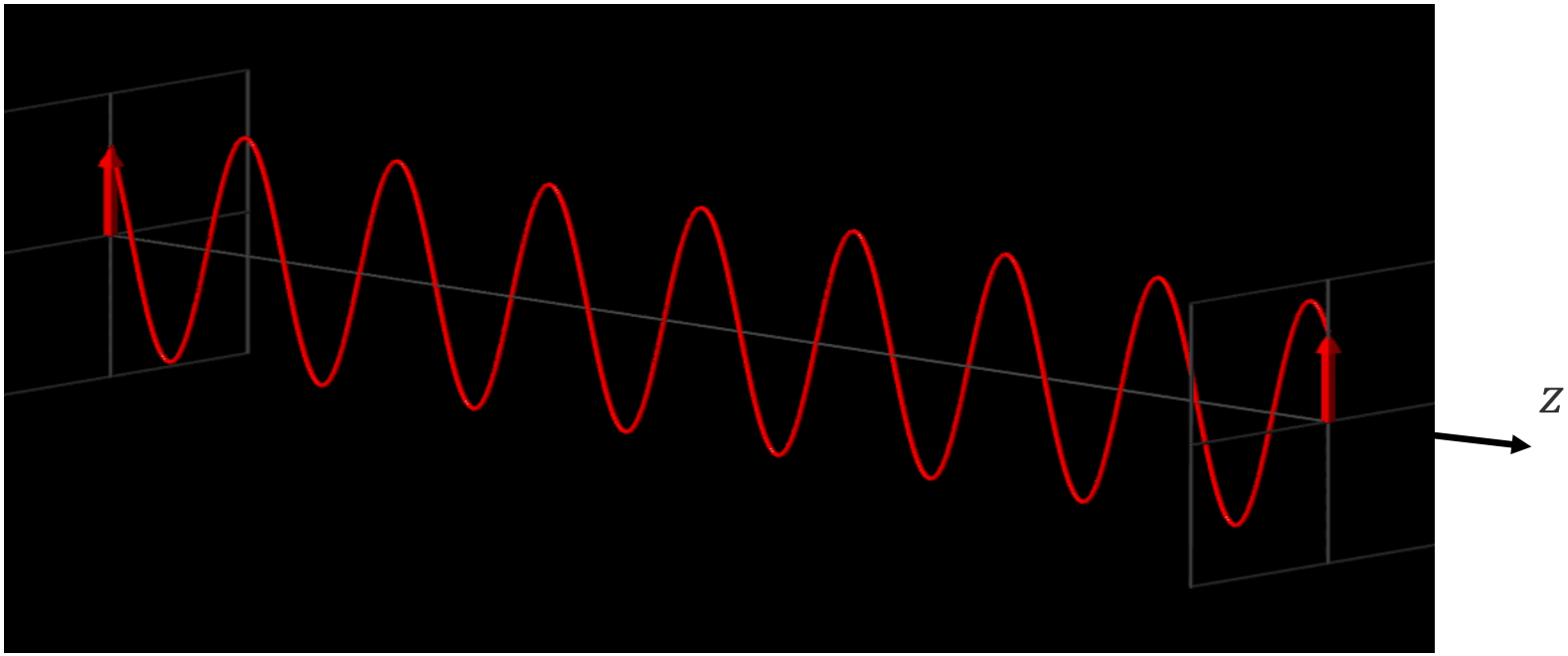
$$\vec{E}(z,t) = E_o \sin(\omega t \pm kz)\ \vec{a}_x$$

Field orientation/polarization

$\omega = 2\pi f$ Frequency

$k = \frac{\omega}{v}$ Wave number

Direction of wave propagation



Q & A

Send me an email if you have any questions.